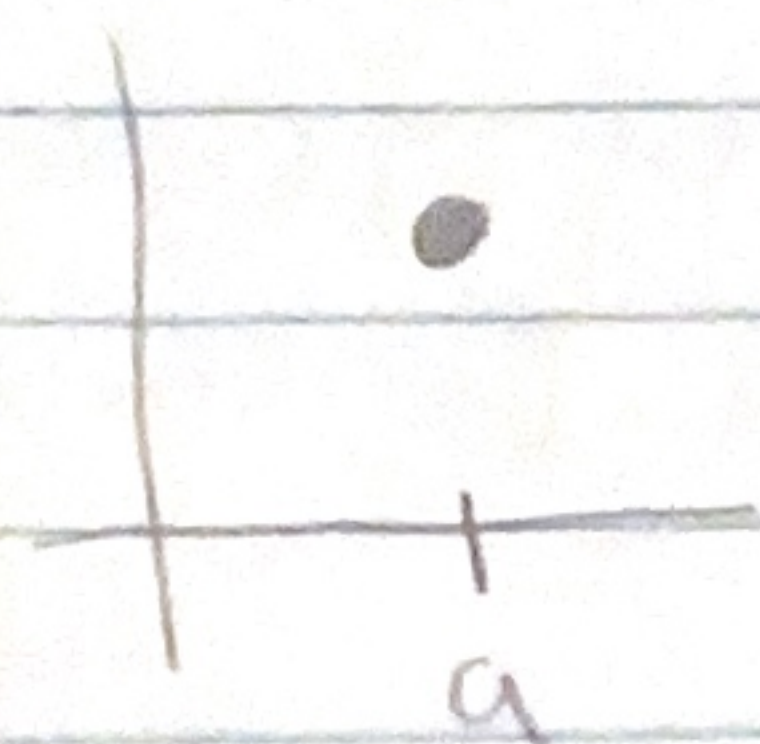


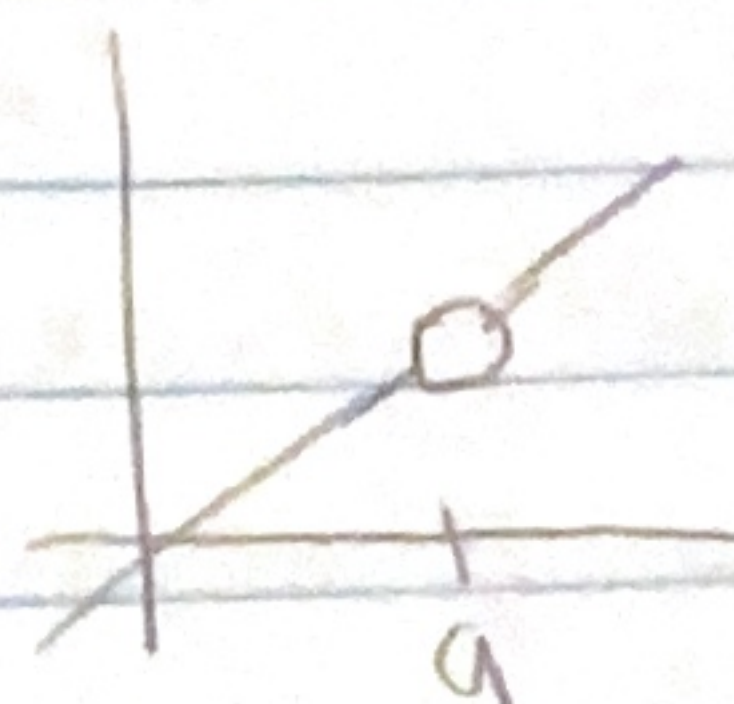
Continuity

For something to be continuous these must be available

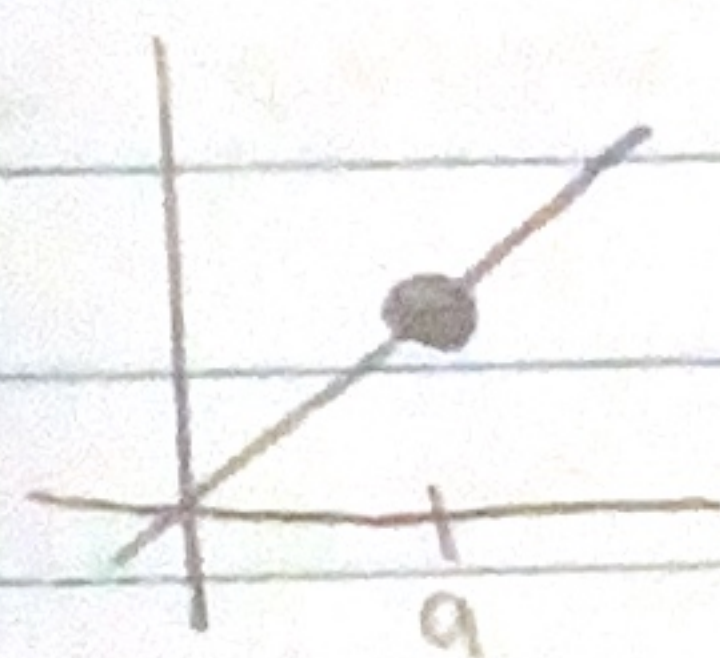
$f(a)$ is defined



$\lim_{x \rightarrow a} f(x)$ exists

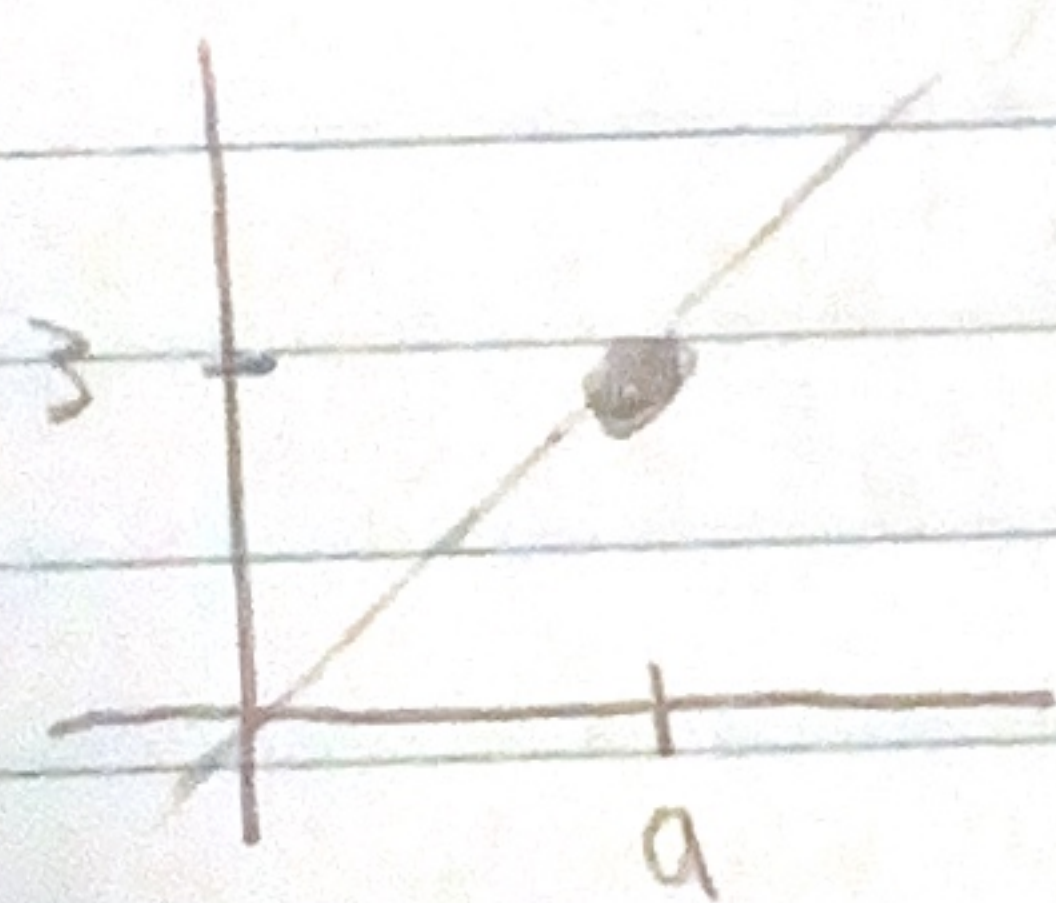


$\lim_{x \rightarrow a} f(x) = f(a)$



if those exist then it is a continuous limit of function
if they do not exist they are discontinuous

example:



$\lim_{x \rightarrow a} f(x) = 3$

$f(a) = 3$

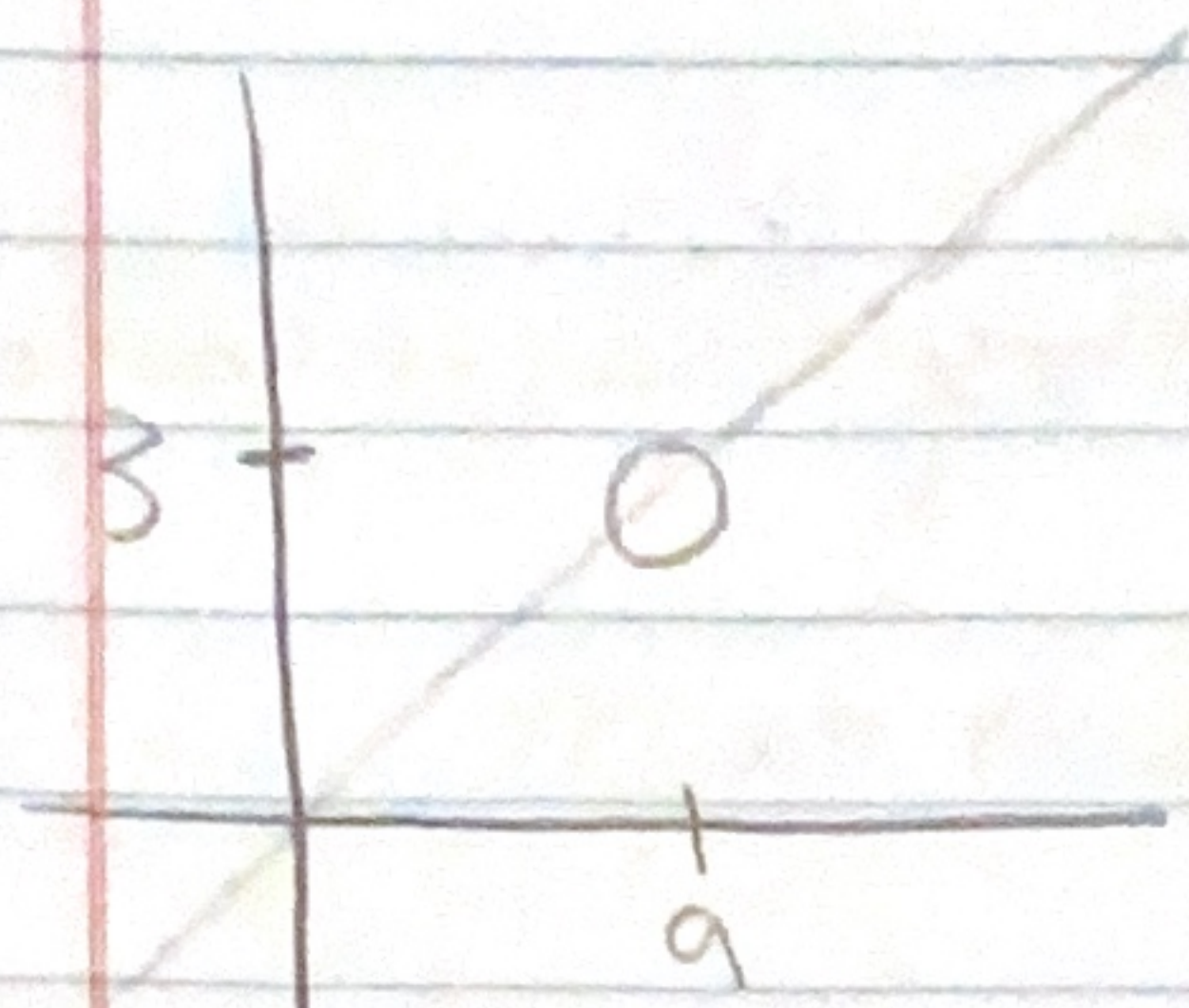
you can see that the
 $\lim_{x \rightarrow a} f(x) = 3 = f(a) = 3$

the answer is the same
for both which fits the
criteria to be continuous

- for a function to be continuous

$\lim_{x \rightarrow a} f(x) = f(a)$

they need to be the same



$$\lim_{x \rightarrow a} f(x) = 3$$

$$f(a) = \text{DNE}$$

this is the same as the last one the only difference is this is not a continuous function

because

$$\lim_{x \rightarrow a} f(x) = 3 \neq f(a) = \text{DNE}$$

they are not the same

for it to be continuous it must be or have the same answer

removable discontinuity or (hole)

for a hole to exist a limit must exist but the function cannot exist

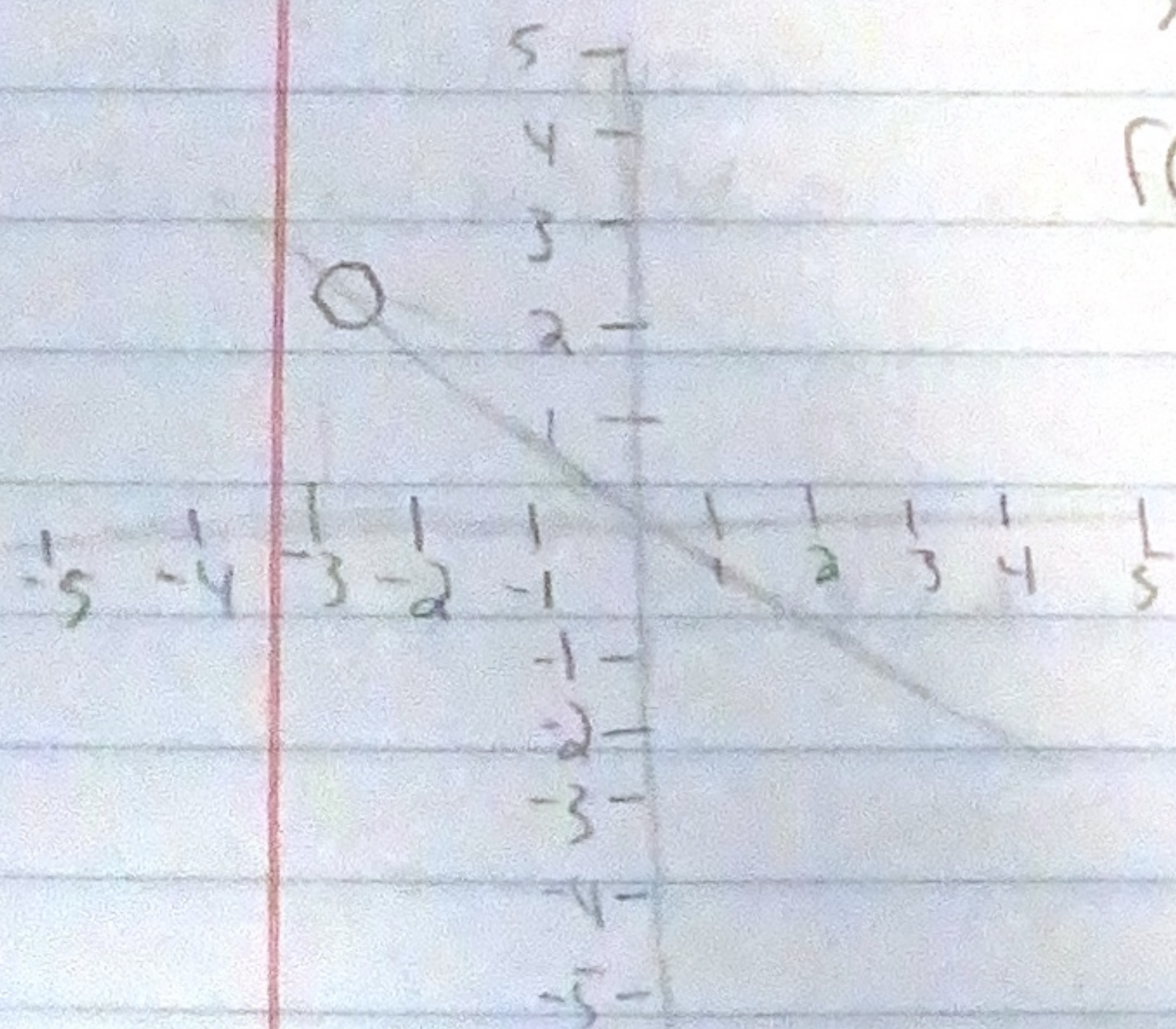
$$\lim_{x \rightarrow a} f(x) \neq f(a) \text{ for hole to exist}$$

example -

$$\lim_{x \rightarrow -3} f(x) = 2 \text{ — not the same —}$$

when make it a hole

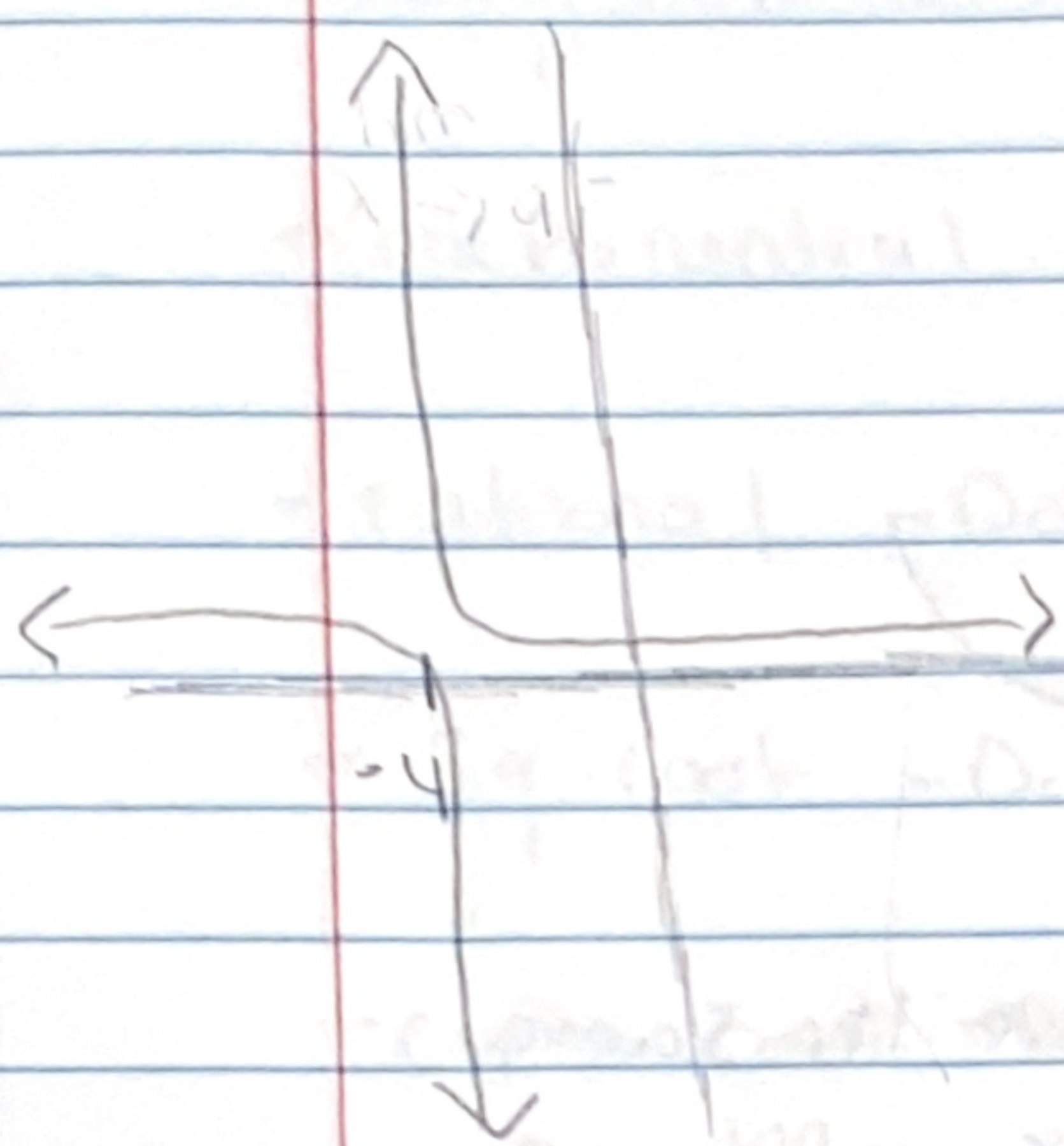
$$f(-3) = \text{DNE}$$



this is what an infinite discontinuity needs to look like

$$\lim_{x \rightarrow a^-} f(x) = \pm \infty \quad \text{or} \quad \lim_{x \rightarrow a^+} f(x) = \pm \infty$$

→ infinite discontinuity (vertical asymptote)



$$\lim_{x \rightarrow a^-} f(x) = \pm \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \pm \infty$$

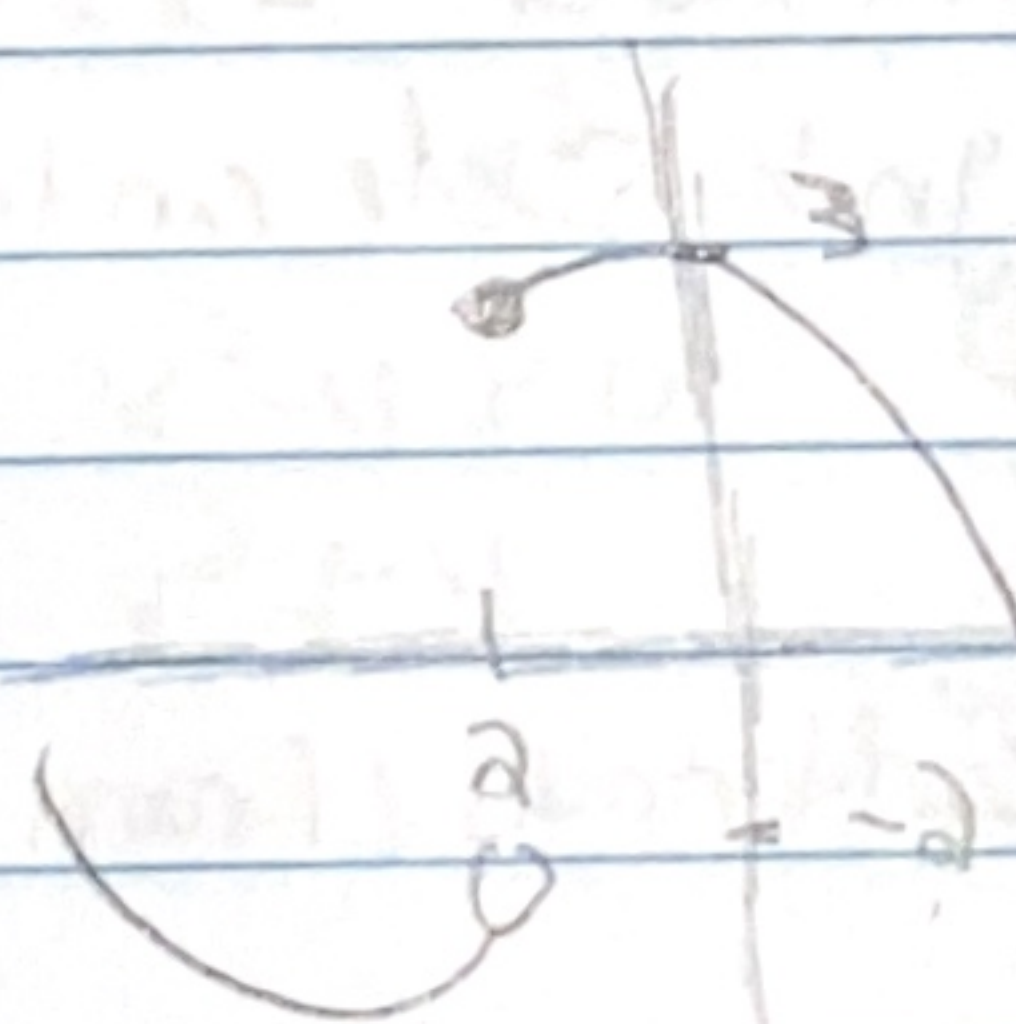
at $x = -4$ there is a infinite discontinuity we know this because

$$\lim_{x \rightarrow -4^-} f(x) = -\infty$$

$$\lim_{x \rightarrow -4^+} f(x) = +\infty$$

looking at the graph at $x = -4$ we have a infinite discontinuity

→ Jump discontinuity (limit dne)



$$\lim_{x \rightarrow 2^+} f(x) = 3$$

$$\lim_{x \rightarrow 2^-} f(x) = -2$$

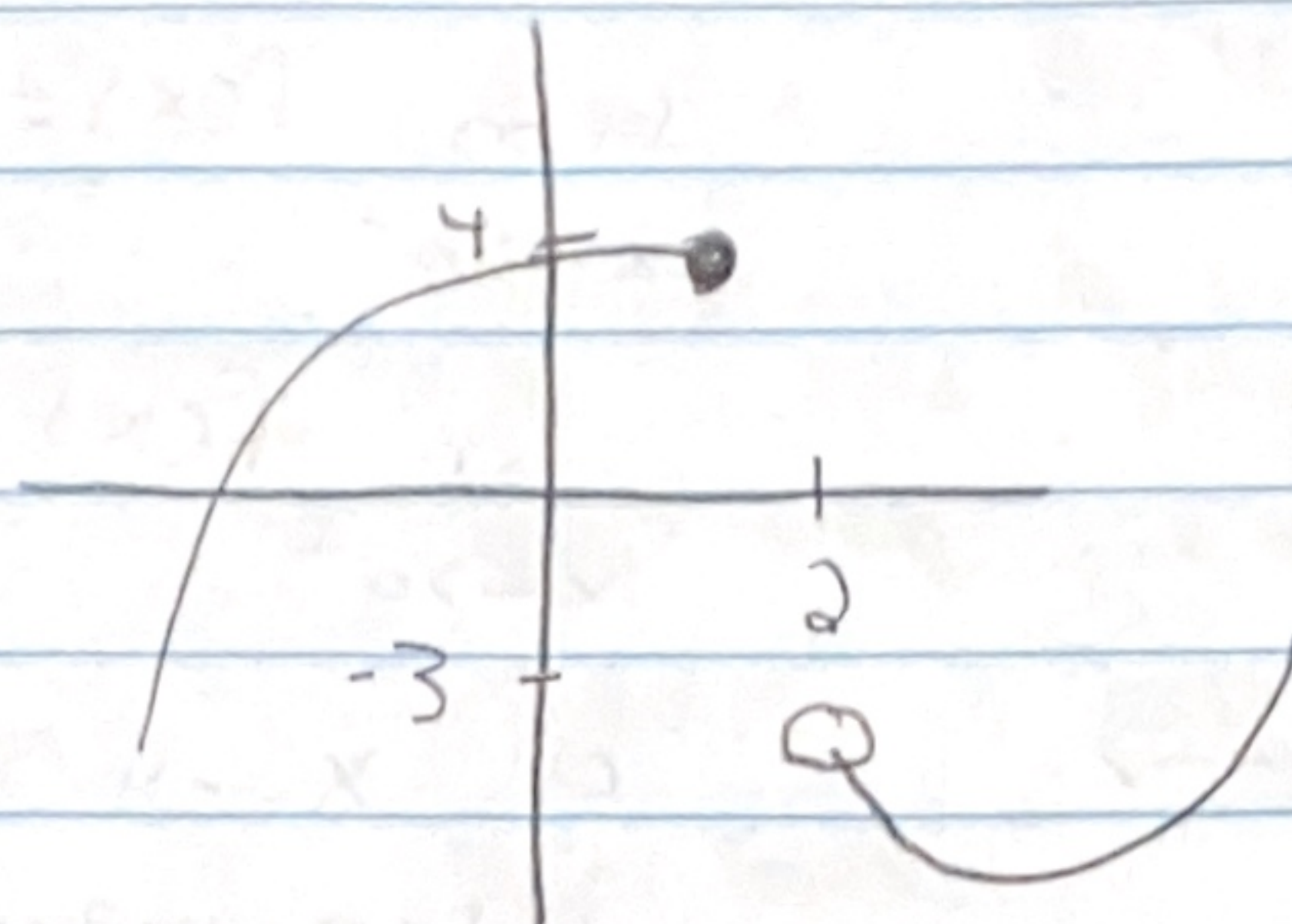
then two limit approaching two are not the same therefore the limit dne

→ a jump discontinuity looks like this

$$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$$

- a function is continuous from the left

$$\lim_{x \rightarrow 2^-} f(x)$$



$$\begin{aligned} \lim_{x \rightarrow 2^-} &= 4 \\ \lim_{x \rightarrow 2^+} &= -3 \\ f(2) &= 4 \end{aligned}$$

- these two have the same answer as x approach 2 from the negative $f(x)$ is 4 and $f(2)$ is 4

this makes it left continuous because the left side of the limit has the same answer as the function the can also work for the right. if x is approaching from the right and it has the same answer as the function then you can call it right side continuous.

- this is what a limit that is continuous from the left should look like

$$\lim_{x \rightarrow a^-} f(x) = f(a) \quad (\text{the same answer})$$

- this is what a limit that is continuous from the right should look like

$$\lim_{x \rightarrow a^+} f(x) = f(a) \quad (\text{the same answer})$$

domain

- Polynomial - Domain $(-\infty, \infty)$ all real #
- exponential - Domain $(-\infty, \infty)$ all real #
- rational - Domain $x > 0$ greater than zero
- Sq root - Domain $x \geq 0$ greater or equal to zero
- cube root - Domain $(-\infty, \infty)$ all real #
- logarithmic - Domain $x > 0$ greater than zero

- the domain is a set of all possible value of x that does not make the function undefined

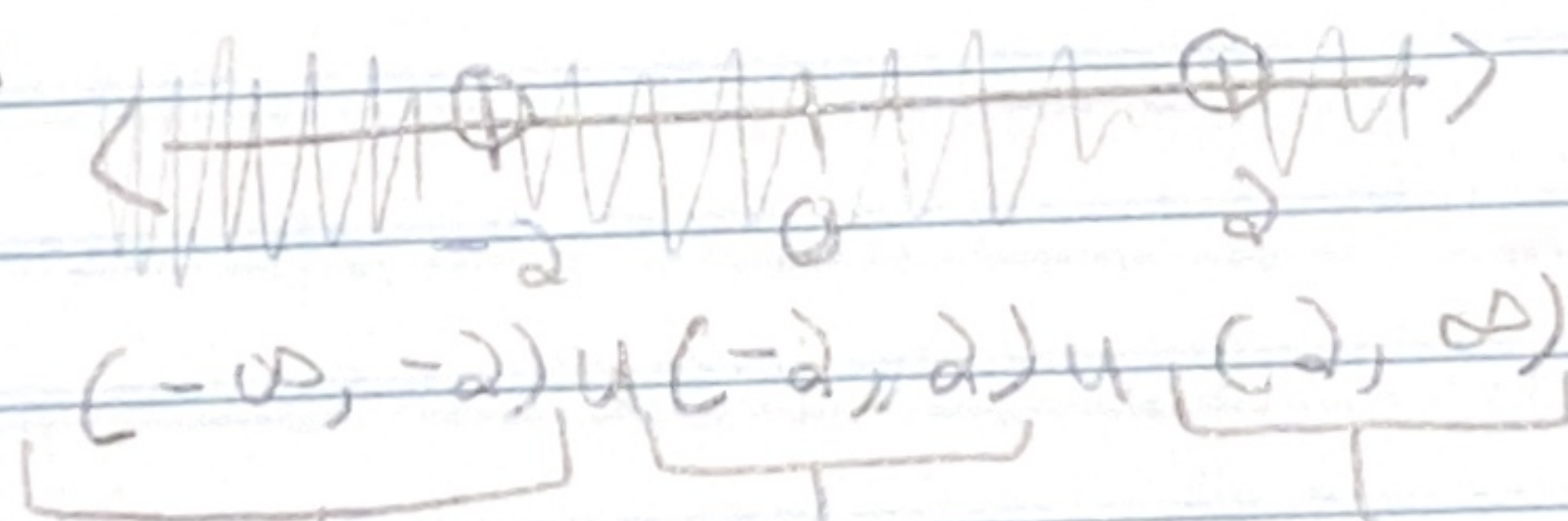
$$f(x) = \frac{x+2}{x^2-4}$$

rational function

$$x^2 - 4 \neq 0$$

$$x^2 \neq 4$$

$$x \neq \pm 2$$



accepts any
negative x-value
except -2

cannot work
or won't be
accepted

does not accept 2
but any positive x-value
works

[bracket] = accepts it

(Parenthesis) = does not accept it